Question 1

Question 1a

For $f(x) = \sqrt{2x+5}$, using the definition of derivative to compute $f'(x)$:

$$\begin{aligned}
f'(x) &= \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} \\
&= \lim_{h\to 0} \frac{\sqrt{2(x+h)+5}-\sqrt{2x+5}}{h} \\
&= \lim_{h\to 0} \frac{\sqrt{2x+2h+5}-\sqrt{2x+5}}{h} \cdot \frac{\sqrt{2x+2h+5}+\sqrt{2x+5}}{\sqrt{2x+2h+5}+\sqrt{2x+5}} \\
&= \lim_{h\to 0} \frac{2x+2h+5-2x-5}{h\left(\sqrt{2x+2h+5}+\sqrt{2x+5}\right)} \\
&= \lim_{h\to 0} \frac{2h}{h\left(\sqrt{2x+2h+5}+\sqrt{2x+5}\right)} \\
&= \lim_{h\to 0} \frac{2}{\sqrt{2x+2h+5}+\sqrt{2x+5}} \\
&= \frac{2}{\sqrt{2x+5}+\sqrt{2x+5}} \\
&= \frac{1}{\sqrt{2x+5}}
\end{aligned}$$

Question 1b

The equation of the tangent line at $x = 2$ is:

$$\begin{aligned}
y - f(2) &= f'(2)(x-2) \\
&= \frac{1}{\sqrt{2(2)+5}}(x-2) + \sqrt{2(2)+5} \\
&= \frac{1}{3}(x-2)+3 \\
&= \frac{1}{3}x + \frac{7}{3}
\end{aligned}$$

Question 2

$F(x) = xf(x) + \sin(g(x))$. Given that:

1. $f(2) = 1$
2. $g(2) = 0$
3. $f'(x) = 2$
4. $g'(x) = 3$

We can find $F'(2)$:

$$\begin{aligned}
F'(2) &= (xf'(x) + f(x)) + (\cos(g(x))g'(x)) \\
&= (2f'(2) + f(2)) + (\cos(g(2))g'(2)) \\
&= (2(2) + 1) + (\cos(0)(3)) \\
&= 8
\end{aligned}$$

Question 3

For $x^3 + y^3 - xy^2 = 5$, $\frac{dy}{dx}$:

$$\begin{aligned}
x^3 + y^3 - xy^2 &= 5 \\
3x^2 + 3y^2\frac{dy}{dx} - (y^2 + 2yx\frac{dy}{dx}) &= 0 \\
3x^2 - y^2 + (3y^2 - 2yx)\frac{dy}{dx} &= 0 \\
3x^2 - y^2 &= (2yx - 3y^2)\frac{dy}{dx} \\
\frac{dy}{dx} &= \frac{3x^2 - y^2}{(2yx - 3y^2)}
\end{aligned}$$

Question 4

$$\begin{aligned}
f(x) &= \sqrt{\frac{x(x+2)}{(2x+1)(3x+2)}} \\
\ln f(x) &= \ln\sqrt{\frac{x(x+2)}{(2x+1)(3x+2)}} \\
&= \frac{1}{2}(\ln(x(x+2)) - \ln((2x+1)(3x+2))) \\
\frac{f'(x)}{f(x)} &= \frac{1}{2}(\frac{2x+2}{x^2+2x} - \frac{12x+7}{6x^2+7x+2}) \\
f'(x) &= \frac{1}{2}\sqrt{\frac{x(x+2)}{(2x+1)(3x+2)}}(\frac{2x+2}{x^2+2x} - \frac{12x+7}{6x^2+7x+2})
\end{aligned}$$

Question 5

The maximum value of $f(x) = x^a(1-x)^b, \quad 0 \leq x \leq 1$ if $a, b > 0$ is:

$$\begin{aligned}
f(x) &= x^a(1-x)^b \\
f'(x) &= ax^{a-1}(1-x)^b + x^a b(1-x)^{b-1}(-1) \\
f'(x) &= ax^{a-1}(1-x)^b - x^a b(1-x)^{b-1}
\end{aligned}$$

We let $f'(x) = 0$ to find the inflection & maxima point between $0, 1$, as we know $f(0) = f(1) = 0$:

$$\begin{aligned} x^a b(1-x)^{b-1} &= a x^{a-1}(1-x)^b \\ xb &= (1-x)a \\ xb &= a - xa \\ x(b+a) &= a \\ x &= \frac{a}{a+b} \\ &\therefore \text{The maximum value is } f(\frac{a}{a+b}) \end{aligned}$$

Question 6

To show that $|\cos x - \cos y| \leq |x - y|$, we can use the Mean Value Theorem:

$$\begin{aligned} \text{Let } f(x) &= \cos x \\ f'(x) &= -\sin x \\ \text{Using the MVT: } f'(c) &= \frac{f(x) - f(y)}{x - y} \\ -\sin c &= \frac{\cos x - \cos y}{x - y} \\ |\cos x - \cos y| &= |x - y||\sin c| \end{aligned}$$

From the last line, as $|\sin c|$ has a domain $[0, 1]$, consider the following cases:

1. $|\sin c| = 1$: $|\cos x - \cos y| = |x - y|$
2. $1 > |\sin c| > 0$: $|\cos x - \cos y| < |x - y|$
3. $|\sin c| = 0, x \neq y$: $|\cos x - \cos y| = 0 < |x - y|$

Therefore, $|\cos x - \cos y| \leq |x - y|$.

Question 7

$$\begin{aligned} \lim_{x\to\infty} (\frac{x}{x+1})^x &= \lim_{x\to\infty} (e^{x \ln(\frac{x}{x+1})}) \\ &= e^{\lim_{x\to\infty}(x \ln \frac{x}{x+1})} \end{aligned}$$

Consider $\lim_{x\to\infty}(x \ln \frac{x}{x+1})$:

$$\begin{aligned} \lim_{x\to\infty} (x \ln \frac{x}{x+1}) &= \lim_{x\to\infty} (\frac{\ln \frac{x}{x+1}}{\frac{1}{x}}) \\ &= \lim_{x\to\infty} (\frac{\ln \frac{1}{1+\frac{1}{x}}}{\frac{1}{x}}) \\ \text{As substituting results in } \frac{0}{0}, \text{ We can apply L'Hopital's Rule: } &= \lim_{x\to\infty} (\frac{(1+\frac{1}{x})\frac{1}{(1+\frac{1}{x})^2}\frac{1}{x^2}}{-\frac{1}{x^2}}) \\ &= \lim_{x\to\infty} (-\frac{1}{1+\frac{1}{x}}) \\ &= -1 \end{aligned}$$

Therefore, the original limit is e^{-1}.

Question 8

For $f(x) = xe^x$

Question 8a

$$\begin{aligned}\lim_{x\to a} xe^x &= \lim_{x\to a} \frac{x}{e^{-x}} \\ &= \lim_{x\to a} \frac{1}{-e^{-x}} \\ &= \frac{1}{-e^{-a}} \neq \pm\infty \ \forall\ a\end{aligned}$$

Hence, there is no vertical asymptote.

$$\begin{aligned}\lim_{x\to\infty} xe^x &= \lim_{x\to\infty} \frac{x}{e^{-x}} \\ &= \lim_{x\to\infty} \frac{1}{-e^{-x}} \\ &= 0\end{aligned}$$

Hence, there is a horizontal asymptote at $y = 0$.

Question 8b

$$\begin{aligned}f'(x) &= e^x + xe^x \\ \text{For } f'(x) &= 0 \\ e^x + xe^x &= 0 \\ \because e^x &> 0 \\ \therefore x &= -1\end{aligned}$$

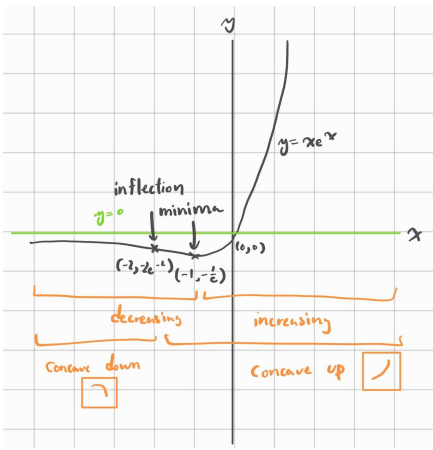
Therefore, there is a critical point at $x = -1$.

$$\begin{aligned}f''(x) &= 2e^x + xe^x \\ f''(-1) &= 2e^{-1} - e^{-1} \\ &= e^{-1} \\ &> 0\end{aligned}$$

Thererfore, there is a local minimum at $(-1, -\frac{1}{e})$.

Question 8c

$$
\begin{aligned}
f''(x) &= 2e^x + xe^x \\
\text{For } f''(x) &= 0 \\
2e^x + xe^x &= 0 \\
e^x(2+x) &= 0 \\
\because e^x &> 0 \\
\therefore x &= -2
\end{aligned}
$$

Therefore, there's a infection point at $(-2, -2e^{-2})$.

Question 8d